



LEVER HOISTS





Specifications for Hoists and Trolleys



Items to Consider When Selecting Powered Hoists and Trolleys

- Capacity (lbs., tons, or kg.)
- Lift (ft. or m)
- Lift speed (ft. / min.) (m / min.)
- Lifting medium (chain or wire rope)
- Power (electric, air, etc.)
- Power cord length
- Push button cord length
- Type of suspension
- Type of trolley
(manual, powered, plain, geared)
- Headroom requirements
- Beam size and type
- Trolley speed
- Trolley brake requirements
- Power Supply voltage
- Control voltage
- Single, two speed or variable speed
- Duty cycle requirements
- Special electrification needs
- Abnormal operating conditions
- Chain container

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The following are trademarks of Columbus McKinnon Corporation: **ShopAir, LodeRail.**

Series 602/603 mini ratchet lever hoist

**Mini-Ratchet Lever Hoists lift up to 1100 lbs.
plus the convenience of toolbox storage.**

The CM Series 603 & 602 Mini- Hoists are the most compact ratchet lever hoists in the market, perform just like the larger models, plus come standard with these great features:

- Plated load chain is standard.
- Weston-type load brake.
- Rubber hand grip for better comfort and security.
- Lightweight design for ease of portability. Use in confined conditions with one-hand operation.
- Forged upper and lower hooks with latches standard.
- Impact resistant, all-steel frame, gear case and cover.
- Hardened link-type alloy steel load chain for strength and long wear.
- Free wheeling feature serves to quickly attach the load.
- Metric rated.



Our compact Mini-Hoists perform just like the larger hoists, except they're small enough to fit in the palm of your hand.



Series 602 Convenient Carry Bag -
Product Code # 0212



Series 603 Convenient Carry Bag -
Product Code # 0217

SPECIFICATIONS

Product Code	Model Number	Rated Capacity (Lbs.) (Kg.)		Standard Lift	Number of Chainfalls	Minimum Distance Between Hooks (In.)	Lift w/ One Full Lever Turn (In.)	Handle Length (In.)	Handle Pull @ W.L.L. (Lbs.)	Approx. shipping wgt. (lbs)
0210	Series 602	550	250	5	1	9.45	3.15	6.300	56	6
0214	Series 602 w/10' lift	550	250	10	1					7
0212	Series 602 Carry Bag									.3
0215	Series 603	1,100	500	5	1	11.125	1.57	6.375	78	7
0216	Series 603 w/10' lift	1,100	500	10	1	11.125	1.57	6.375	78	8
0217	Series 603 Carry bag									.3

Dimensional Drawing - 602



Bandit Ratchet Lever Hoist

The CM® Bandit™ is one of the most compact and durable ratchet lever hoists in the industry. Its lightweight and portable design, easy free-chaining feature and 360° rotating handle make it one of the most versatile hoists on the market today. Now available in 3/4, 1-1/2, 3 and 6 ton capacities for all of your pulling and lifting needs.

CAPACITIES & LIFT

Capacities of 3/4, 1-1/2, 3 & 6 tons. Metric Rated. Standard lifts up to 20 feet. Longer lifts are available.

COMPACT & PORTABLE DESIGN

Rugged, yet lightweight, design and construction allow the Bandit to be easily transported and used in even the most confining spaces.

OUR FIRST HOIST WITH

CM® SMART ID™ RFID TECHNOLOGY!

- Standard on all U.S. Bandit units.
- Chip associated with detailed product information at the factory.



FREE-CHAINING

Easy free-chaining feature allows for quick take up and positioning of slack chain. Designed not to accidentally free chain while under load.

360° ROTATING HANDLE

Full rotation of handle allows for versatile rigging options when working in tight spaces.

LOW HANDLE EFFORT

Double reduction gearing provides easy operation with minimal handle effort.

COMFORT-FIT HANDLE

Screwed-on comfortable rubber grip makes for a secure hold in all environments.

LOAD SECURITY & EASY INSPECTION

Upper and lower hooks feature standard cast safety latches that provide positive and secure load engagement. Hooks are bolted on for easy removal and inspection.

MEETS ASME B30.21

POSITIVE LOAD CONTROL

Enclosed Weston-type load brake stays clean and dry for positive load positioning.

EASY TO POSITION & ATTACH

Upper and lower hooks feature extra-wide throat openings to allow for easier attachment to pick points. Hooks swivel 360° for faster positioning.

BUILT TO LAST

Impact-resistant stamped steel housing withstands repeated rigorous use.

CORROSION RESISTANT

Standard powder coat finish on housing and zinc plating on major components for extra protection against corrosion when working in harsh environments.

100% LOAD TESTED GOLD CHROMATE CHAIN

Proven chain integrity and corrosion resistance.

EASY IDENTIFICATION

Each unit has a unique serial number for easy and accurate identification.

OPTIONAL SHIPYARD HOOKS

Shipyards hooks are available as an option on 1-1/2 & 3 ton units.

OPTIONAL INTERNAL LOAD LIMITER

Allows the handle to rotate but will not lift if load exceeds 125% of the working load limit.

BANDIT HOISTS SHIP IN 3 DAYS OR LESS – GUARANTEED

3/4 & 1-1/2 ton units are eligible for our In-Stock Guarantee.

FAST & EASY PARTS ACCESS

Genuine CM Bandit parts are available in Total Repair Kits and convenient bulk packaging.



3/4 ton US model pictured above

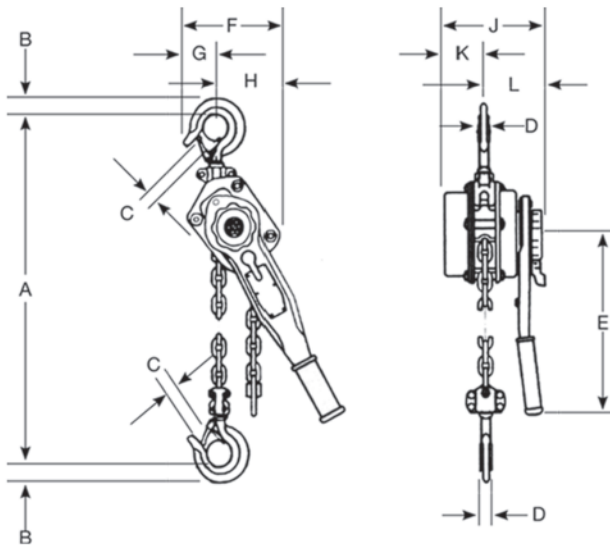


THE WORLD'S FIRST
HMI-CERTIFIED
RATCHET LEVER HOIST

Bandit Ratchet Lever Hoist product at a glance

Capacities (Ton)	In-Stock Guarantee	CM Smart ID RFID	Initial Assembly	Final Assembly	Hook Origin	Tri-Point Hook Inspection	Optional Shipyard Hooks	Chain Origin	Type of Chain	Load Tested
3/4	✓	✓	CMCO China Facility	CMCO USA Facility	CMCO USA Facility			CMCO USA Facility	Gold Chromate	China & USA
1-1/2	✓	✓			CMCO USA Facility		✓			
3		✓			CMCO China Facility	✓	✓			
6		✓			CMCO China Facility	✓				

Check for availability on 3 and 6 ton units.



BANDIT TRI-POINT HOOK INSPECTION

Load hooks feature 3 raised points that allow you to measure the hook to detect any indication of overload or deformation.

Available on 3 & 6 ton capacity models.



Bandit Ratchet Lever Hoist specifications

Model Number *	Capacity Metric Rated Ton (kg)	Standard Lift FT (m)	Lever Pull to Lift Rated Load LBF (kg)	Net Weight LB (kg)	Dimensions Inches (mm)										
					A	B	C	D	E	F	G	H	J	K	L
BAN07505	3/4 (750)	5 (1.52)	45 (20.41)	13.6 (6.17)	12.80 (325.12)	0.94 (23.88)	1.13 (28.70)	0.69 (17.53)	9.63 (244.60)	4.88 (123.95)	2.19 (55.63)	3.31 (84.07)	6.06 (153.92)	2.11 (53.59)	3.95 (100.33)
BAN07510		10 (3.05)		15.7 (7.12)											
BAN07515		15 (4.57)		18 (8.16)											
BAN07520		20 (6.1)		20.1 (9.12)											
BAN15005	1-1/2 (1500)	5 (1.52)	64 (29.03)	20.7 (9.39)	14.20 (360.68)	1.13 (28.70)	1.25 (31.75)	0.81 (20.57)	10.25 (260.35)	5.63 (143.00)	2.38 (60.45)	3.56 (90.42)	6.75 (171.45)	2.69 (68.33)	4.13 (104.90)
BAN15010		10 (3.05)		24.3 (11.02)											
BAN15015		15 (4.57)		27.9 (12.65)											
BAN15020		20 (6.1)		31.5 (14.29)											
BAN30005	3 (3000)	5 (1.52)	73 (33.11)	38.3 (17.37)	17.50 (444.5)	1.69 (42.93)	1.56 (39.62)	1.25 (31.75)	16.37 (415.8)	7.52 (191.01)	3.38 (85.85)	4.65 (118.11)	7.84 (199.14)	3.29 (83.57)	4.56 (115.82)
BAN30010		10 (3.05)		45.7 (20.73)											
BAN30015		15 (4.57)		53 (24.04)											
BAN30020		20 (6.1)		60.4 (27.4)											
BAN60005	6 (6000)	5 (1.52)	93 (42.18)	63.2 (28.67)	22.00 (558.8)	1.81 (45.97)	2.19 (55.63)	1.50 (38.1)	16.37 (415.8)	9.47 (240.54)	3.38 (85.85)	6.05 (153.67)	7.84 (199.14)	3.29 (83.57)	4.56 (115.82)
BAN60010		10 (3.05)		77.9 (35.33)											
BAN60015		15 (4.57)		92.6 (42.00)											
BAN60020		20 (6.1)		107.3 (48.7)											

Check for availability on 3 and 6 ton units.

Tornado 360 Ratchet Lever Hoist

Redefining lever-operated hoists, the CM® Tornado 360™ features the revolutionary Sidewinder™ lever handle that allows for efficient operation in both lifting and pulling applications. Ergonomically designed for increased safety, the patent-pending CM Tornado 360° lets the operator work up to 12 times faster and with as much as 30% less pull force than with conventional ratchet lever tools.

Available in 3/4, 1-1/2, 3 and 6 Ton capacities with standard lifts up to 20 feet.

STANDARD FEATURES

SIDEWINDER LEVER HANDLE

This first-of-its-kind lever handle design will revolutionize the ratchet lever hoist industry. Its unique foldable handle and 360° rotating lever increases productivity while reducing the risk of operator injury.

CONVENIENT DIRECTIONAL INDICATOR

Easy-to-use, highly visible directional indicator located on the handle clearly shows the operating direction as lifting, lowering or neutral.

EASY & SMOOTH FREE CHAINING

For quick take up and positioning of slack chain – even with one hand. Designed not to accidentally free chain while under load. Cast chain end stop allows the user to easily position the chain in free-chaining mode and prevents it from entering the liftwheel and jamming the hoist.

POSITIVE LOAD CONTROL

Enclosed Weston-type brake stays clean and dry for precise load positioning.

BUILT TO LAST

Lightweight aluminum housing withstands rigorous use and features high-quality, long-lasting bearings. Powder coat finish provides extra protection in harsh environments.

MEETS ASME B30.21

EXCELLENT CORROSION PROTECTION

Robust chain guide and chain stripper made from cast steel and zinc-plated for corrosion protection.

100% LOAD TESTED CHAIN

U.S. units contain gold chromate chain designed and manufactured in the U.S.A. by Columbus McKinnon. Canadian units contain zinc-plated chain. Both chain finishes protect against corrosion.

EASY-TO-INSPECT SWIVEL HOOKS

Bolt-on hooks with nylon locks help users comply with ASME B30.21 and B30.10 inspection requirements. Hooks are forged, allowing them to yield under overload without breaking. Cast safety latches provide positive and secure load engagement.

INDUSTRY-LEADING WARRANTY

Our lifetime warranty on all mechanical components, including the Sidewinder lever handle, is the industry's best warranty against manufacturing and material defects.

With our 3-year brake warranty, if the brake discs wear out within 3 years from the date of purchase, CMCO will replace the ratchet disc assembly free of charge.

EASY IDENTIFICATION

Each unit has a unique serial number for easy and accurate identification.



US model pictured above

OPTIONAL FEATURES

SHIPYARD HOOKS

Available on 1-1/2 and 3 ton units only.

INTERNAL LOAD LIMITER

Allows the handle to rotate freely, preventing the lifting of an overload sufficient to damage the hoist. Units equipped with a load limiter feature a black hand wheel for quick and easy on-site identification.

CM SMART ID™ RADIO FREQUENCY IDENTIFICATION

Simplify inventory and inspection management processes.

At a Glance

EASILY CHANGE YOUR GRIP TO SUIT THE APPLICATION

The CM Tornado 360° gives operators ultimate flexibility. When an application requires a customary ratcheting hand grip, the Tornado 360° can be operated like a traditional lever tool when the lever handle is in the upright position. inspection management processes.

ERGONOMIC DESIGN DELIVERS OPTIMAL OPERATOR SAFETY

- Hoist design allows the operator to work in a safe and ergonomic position.
- Sidewinder lever handle enables smooth 360° operation, reducing repetitive wrist action experienced with traditional ratchet lever hoists.
- Double reduction gearing and high-quality bearings decrease required pull force by up to 30%, reducing operator fatigue.

EASILY CARRY & TRANSPORT WITH JUST ONE HAND

When folded out, the lever handle can serve as a carrying grip for easy transport. Simply attach the end stop to the hook and slip the hook onto the Sidewinder lever handle to lift and carry the unit with one hand.

EASIER MAINTENANCE & INSPECTION

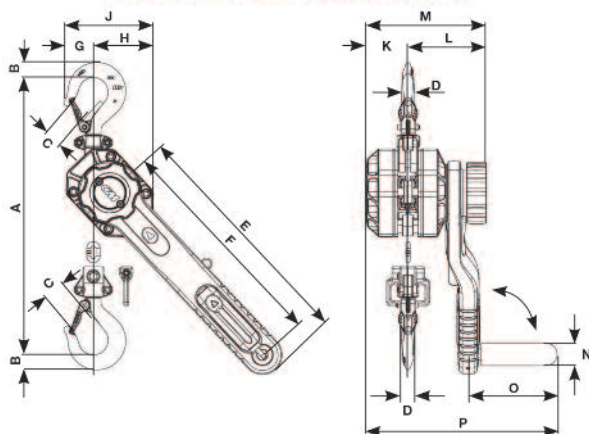
For quicker maintenance and lower repair costs, screws are set into the hoist body to protect them from potential damage.

To ensure optimal performance and safety, demand genuine CMCO parts. Protect your investment, maintain warranty requirements and preserve the integrity of your hoist's superior design – ask for Parts Star™ by Columbus McKinnon.

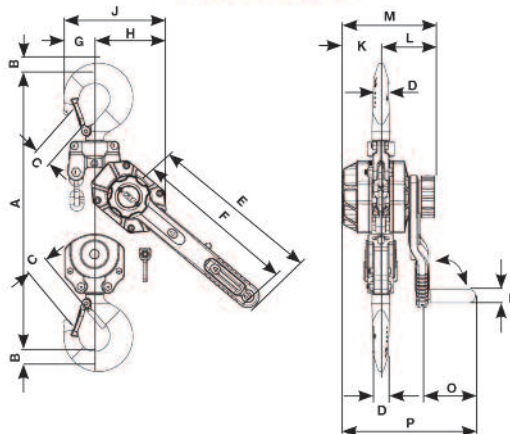
Specifications

Capacity Tons (kg.)	Product Code		Standard Lift ft. (m)	Pull Force to Lift Rated Load lb./ft. (kg./ft.)		Reeving	Weight lbs. (kg)	Dimensions in. (mm)														
	without load limiter	with load limiter		Sidewinder Lever Handle Operation	Traditional Lever Operation			A (Min.)	B	C	D	E	F	G	H	J	K	L	M	N	O	P
0.75 (750)	TOR07505	TOR07505LL	5 (1.5)	38.2 (17.3)	47.2 (21.4)	1	14.6 (6.62)	12.60 (320)	0.79 (20)	1.06 (27)	0.71 (18)	12.87 (327)	11.81 (300)	1.57 (40)	3.19 (81)	4.76 (121)	2.20 (56)	4.13 (105)	6.33 (161)	1.18 (30)	4.72 (120)	10.1 (257)
	TOR07510	TOR07510LL	10 (3.0)				16.8 (7.62)															
	TOR07515	TOR07515LL	15 (4.5)				19.0 (8.62)															
	TOR07520	TOR07520LL	20 (6.0)				21.2 (9.62)															
	TOR075LCX	TOR075LCXLL	Less Chain				11.95 (5.42)															
1.5 (1,500)	TOR15005	TOR15005LL	5 (1.5)	58.5 (26.5)	60.7 (27.5)	1	21.0 (9.52)	14.76 (375)	1.02 (26)	1.22 (31)	0.83 (21)	12.87 (327)	11.81 (300)	2.01 (51)	3.78 (96)	5.79 (147)	2.71 (69)	4.33 (110)	7.05 (179)	1.18 (30)	4.72 (120)	10.7 (273)
	TOR15010	TOR15010LL	10 (3.0)				24.5 (11.11)															
	TOR15015	TOR15015LL	15 (4.5)				28.1 (12.75)															
	TOR15020	TOR15020LL	20 (6.0)				31.6 (14.33)															
	TOR150LCX	TOR150LCXLL	Less Chain				16.71 (7.58)															
	00002145	Optional Upper Shipyard Hook					4.4 (2.00)	14.02 (356)	1.10 (28)	1.06 (27)	0.91 (23)	12.87 (327)	11.81 (300)	1.46 (37)	3.78 (96)	5.24 (133)	2.72 (69)	4.33 (110)	7.05 (179)	1.18 (30)	4.72 (120)	10.75 (273)
3315WLP	Optional Lower Shipyard Hook					2.87 (1.30)																
3 (3,000)	TOR30005	TOR30005LL	5 (1.5)	79.4 (36)	90.4 (41)	1	36.61 (16.61)	17.52 (445)	1.46 (37)	1.57 (40)	1.10 (28)	14.84 (377)	13.78 (350)	2.24 (57)	4.84 (123)	7.09 (180)	3.39 (86)	4.76 (121)	8.15 (207)	1.18 (30)	4.72 (120)	11.77 (299)
	TOR30010	TOR30010LL	10 (3.0)				44.01 (19.96)															
	TOR30015	TOR30015LL	15 (4.5)				51.41 (23.32)															
	TOR30020	TOR30020LL	20 (6.0)				58.81 (26.68)															
	TOR300LCX	TOR300LCXLL	Less Chain				29.21 (13.25)															
	00002383	Optional Upper Shipyard Hook					4.23 (1.92)	17.17 (436)	1.46 (37)	1.42 (36)	1.10 (28)	14.84 (377)	13.78 (350)	1.89 (48)	4.84 (123)	6.73 (171)	3.39 (86)	4.76 (121)	8.15 (207)	1.18 (30)	4.72 (120)	11.77 (299)
3403WLP	Optional Lower Shipyard Hook					4.81 (2.18)																
6 (6,000)	TOR60005	TOR60005LL	5 (1.5)	83.8 (38)	92.6 (42)	2	52.26 (23.70)	22.17 (563)	1.77 (45)	1.73 (44)	1.38 (35)	14.84 (377)	13.78 (350)	2.80 (71)	6.38 (162)	9.17 (233)	3.39 (86)	4.76 (121)	8.15 (207)	1.18 (30)	4.72 (120)	11.77 (299)
	TOR60010	TOR60010LL	10 (3.0)				59.66 (27.06)															
	TOR60015	TOR60015LL	15 (4.5)				67.06 (30.42)															
	TOR60020	TOR60020LL	20 (6.0)				74.46 (33.77)															
	TOR600LCX	TOR600LCXLL	Less Chain				44.86 (20.35)															

3/4, 1-1/2 AND 3 TON UNITS



6 TON UNITS



Tornado 360 Ratchet Lever Hoist

THE REVOLUTIONARY SIDEWINDER™ LEVER HANDLE

The patent-pending CM Tornado 360° ratchet lever hoist features the revolutionary Sidewinder™ lever handle. What makes the Sidewinder so unique are two innovative components:

1. 360° ROTATING LEVER

The unique offset, or curved, design of the lever keeps the operator's hand and body aligned with the load chain. This reduces the risk of the twist effect (when a hoist twists around the chain) and the need to stabilize the hoist with another hand.

2. FOLD-OUT REVOLVING HANDLE

Recessed in the lever, the heavy-duty steel core handle folds down into position on either side of the lever, allowing for easy operation from any angle. The handle is grooved to provide a secure grip – even with gloves.

The handle revolves, allowing the operator to ergonomically and easily maintain control of the lever through its full 360° range of motion.

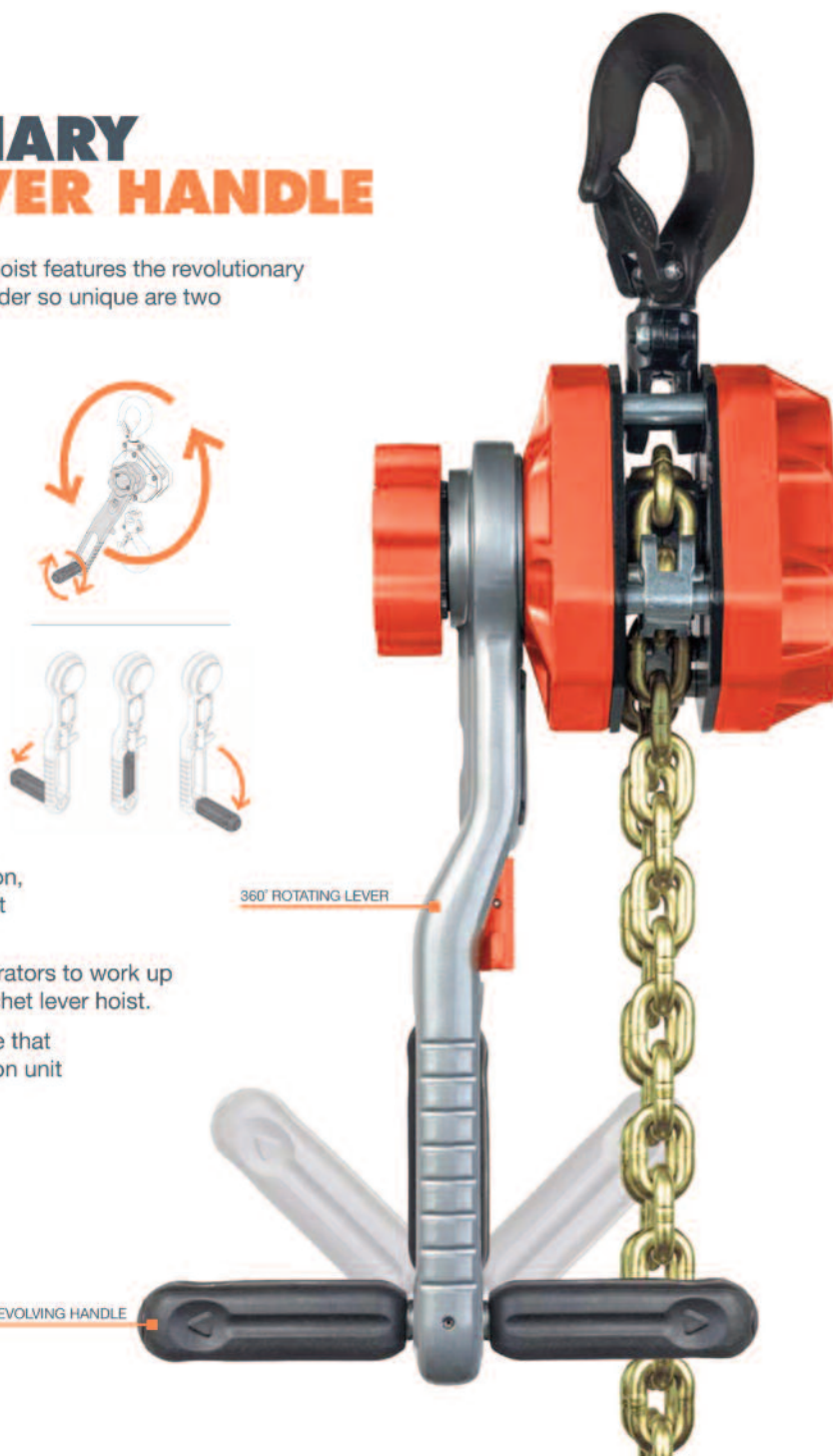
Together, the Sidewinder's lever and handle allow the operator to realize the full potential of 360° rotation, making the CM Tornado 360° unlike any other ratchet lever hoist on the market:

- 360° rotation increases efficiency, allowing operators to work up to 12 times faster than with a conventional ratchet lever hoist.
- Minimal effort with best-in-class lever pull force that beats the competition: 38.2 lb./ft. for the 3/4 ton unit and 58.5 lb./ft. for the 1-1/2 ton unit.

**DISTINCTIVE
CLICKING SOUND**
ENSURES THE HANDLE
IS LOCKED INTO POSITION

To return handle to upright position,
simply pull the handle outward
and snap into place inside the lever.

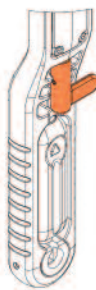
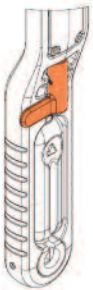
FOLD-OUT REVOLVING HANDLE



EASY-TO-USE DIRECTIONAL CONTROL & FREE CHAINING

CONVENIENT DIRECTIONAL INDICATOR

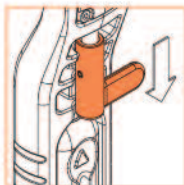
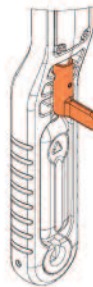
Easy-to-use, highly visible directional indicator window located in the lever clearly shows the operating direction as lifting (▲), lowering (▼) or neutral (N).



EASY & SMOOTH FREE CHAINING

Allows for quick take up and positioning of slack chain – even with one-handed use. Designed not to accidentally free chain while under load. Cast chain end stop allows the user to quickly position the chain in free-chaining mode and prevents it from entering the liftwheel and jamming the hoist.

In this mode of operation, the chain can be pulled through the hoist in either direction by hand for quick attachment to the load. To engage the free chaining feature, remove the load and move the directional lever to the neutral (N) position.



SAFE & SECURE SELECTOR LEVER LOCKS IN PLACE TO PREVENT ACCIDENTALLY SWITCHING.

Pull down on the selector lever to unlock it, turn it to the desired direction, and release it into the locking position.

Series 653 Ratchet Lever hoist

As part of our commitment to bring best-in-class products to market, we are continuously looking to improve existing products. Our popular CM Series 653 ratchet lever hoist has been redesigned with a shorter handle. This modest modification makes a world of difference. Ideal for confined work areas, the Series 653 3/4 and 1-1/2 ton models are now some of the most nimble and easiest to operate hoists on the market today.

CAPACITIES & LIFT

3/4 & 1-1/2 Ton capacities (new shorter handle). 3 & 6 Ton capacities (standard handle). Metric Rated. Standard Lift up to 20 feet.

COMPACT & PORTABLE DESIGN

Take this rugged hoist anywhere. Easily transported and rigged even in the smallest work spaces.

FREE CHAINING

Single-handed free chaining feature allows for quick take up and positioning of slack chain. Designed not to accidentally free chain while under load.

360° ROTATING HANDLE

Full rotation of handle allows for versatile rigging options when working in confined spaces.

RUGGED DESIGN

Impact resistant stamped steel frame and gear cover withstands repeated rigorous use.

BUILT TO LAST

Zinc plated hardened steel chain provides strength and longer wear life which translates to less down time.

POSITIVE LOAD CONTROL

Enclosed Weston type load brake stays clean and dry for positive load positioning.

LOAD CONTROL

Double pawl design ensures that one pawl will be engaged at all times for reliable load control.

SAFE & SECURE

Standard heavy duty forged safety latches on hooks provide positive and secure load engagement.

CORROSION RESISTANT

Standard powder coat finish for extra protection against corrosion when working in harsh environments.

NON-SLIP RUBBER HANDLE GRIP

Comfortable rubber grip provides for extra protection against slippage.

Meets ASME B30.21

5 YEAR WARRANTY

THE ADVANTAGES OF THE SHORTER HANDLE

LESS WEIGHT TO TRANSPORT

Reduces the overall weight of the hoist, making it even easier to carry and operate.

FITS IN SMALL WORK AREAS

The smaller handle allows the Series 653 to fit into confined work areas.

MAKE THE FULL ROTATION

Take full advantage of the 360° rotating handle with less worry of obstructions.

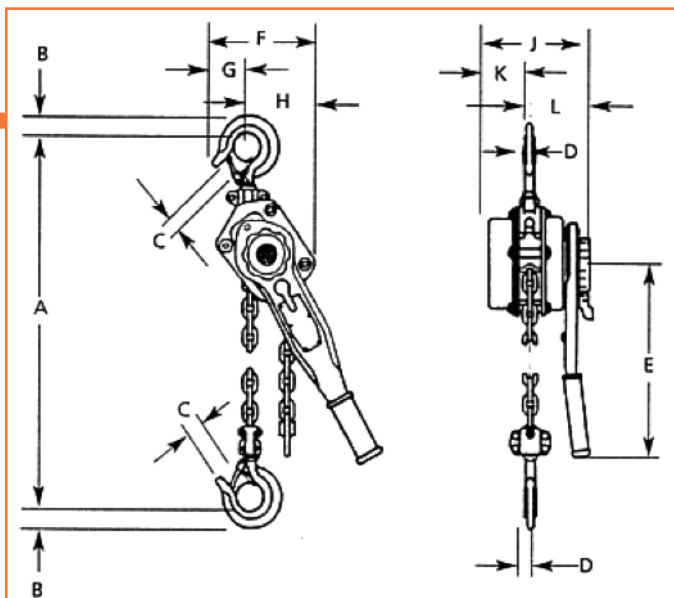


***WARNING**

Overloading and improper use can result in injury.

To avoid injury:

- Do not exceed working load limit, load rating capacity
- Do not use to lift people or loads over people.
- use only alloy chain for overhead lifting.
- Read and follow all instructions.



Series 653 lever operated hoist-specifications



Upper
Shipyard
hook

Capacity (tons)	Product code	Net Wt. (lbs.)
1 1/2	3315WLP	2.9
3	3303WLP	4.9



Lower
Shipyard
hook

1 1/2	3415WLP	2.1
3	3403WLP	4.8

Specifications

Product code	Load rating (tonnes)	Lift or reach (ft.)	Lever pull to lift rated load (lbs.)	Approx. ship wt. (lbs.)	A	B	C	D	E	F	G	H	J	K	L
5310	3/4	5	33	15.0	12 ⁵ / ₈	15/16	1 ¹ / ₈	11/16	9 ¹ / ₄	4 ³ / ₈	2 ³ / ₁₆	2 ³ / ₁₆	5 ¹⁵ / ₁₆	2 ³ / ₈	3 ⁹ / ₁₆
5311	3/4	10	33	15.8	12 ⁵ / ₈	15/16	1 ¹ / ₈	11/16	9 ¹ / ₄	4 ³ / ₈	2 ³ / ₁₆	2 ³ / ₁₆	5 ¹⁵ / ₁₆	2 ³ / ₈	3 ⁹ / ₁₆
5312	3/4	15	33	16.6	12 ⁵ / ₈	15/16	1 ¹ / ₈	11/16	9 ¹ / ₄	4 ³ / ₈	2 ³ / ₁₆	2 ³ / ₁₆	5 ¹⁵ / ₁₆	2 ³ / ₈	3 ⁹ / ₁₆
5313	3/4	20	33	17.3	12 ⁵ / ₈	15/16	1 ¹ / ₈	11/16	9 ¹ / ₄	4 ³ / ₈	2 ³ / ₁₆	2 ³ / ₁₆	5 ¹⁵ / ₁₆	2 ³ / ₈	3 ⁹ / ₁₆
5328	1	5	44	15.0	12 ⁵ / ₈	15/16	1 ¹ / ₈	11/16	9 ¹ / ₄	4 ³ / ₈	2 ³ / ₁₆	2 ³ / ₁₆	5 ¹⁵ / ₁₆	2 ³ / ₈	3 ⁹ / ₁₆
5315	1 1/2	5	51	27.0	14 ³ / ₁₆	1 ¹ / ₈	1 ¹ / ₄	13/16	13	4 ³ / ₈	2 ³ / ₈	2 ³ / ₈	6 ⁷ / ₈	3	3 ¹⁵ / ₁₆
5316	1 1/2	10	51	31.4	14 ³ / ₁₆	1 ¹ / ₈	1 ¹ / ₄	13/16	13	4 ³ / ₈	2 ³ / ₈	2 ³ / ₈	6 ⁷ / ₈	3	3 ¹⁵ / ₁₆
5317	1 1/2	15	51	35.8	14 ³ / ₁₆	1 ¹ / ₈	1 ¹ / ₄	13/16	13	4 ³ / ₈	2 ³ / ₈	2 ³ / ₈	6 ⁷ / ₈	3	3 ¹⁵ / ₁₆
5318	1 1/2	20	51	40.2	14 ³ / ₁₆	1 ¹ / ₈	1 ¹ / ₄	13/16	13	4 ³ / ₈	2 ³ / ₈	2 ³ / ₈	6 ⁷ / ₈	3	3 ¹⁵ / ₁₆
5329	2	5	68	27.0	14 ¹³ / ₁₆	1 ¹ / ₈	1 ¹ / ₄	13/16	16 ¹ / ₄	4 ³ / ₈	2 ³ / ₈	2 ³ / ₈	6 ⁷ / ₈	3	3 ¹⁵ / ₁₆
5320	3	5	77	45.0	18 ¹¹ / ₁₆	1 ¹³ / ₁₆	19/16	1 ¹ / ₄	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆
5321	3	10	77	52.3	18 ¹¹ / ₁₆	1 ¹³ / ₁₆	19/16	1 ¹ / ₄	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆
5330	6	5	77	78.0	23 ¹ / ₄	1 ¹³ / ₁₆	2 ³ / ₁₆	1 ¹ / ₂	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆
5331	6	10	77	94.0	23 ¹ / ₄	1 ¹³ / ₁₆	2 ³ / ₁₆	1 ¹ / ₂	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆
5332	6	15	77	110.0	23 ¹ / ₄	1 ¹³ / ₁₆	2 ³ / ₁₆	1 ¹ / ₂	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆
5333	6	20	77	126.0	23 ¹ / ₄	1 ¹³ / ₁₆	2 ³ / ₁₆	1 ¹ / ₂	16 ¹ / ₄	7 ¹ / ₂	3 ³ / ₈	4 ³ / ₁₆	7 ⁷ / ₈	3 ³ / ₈	4 ⁹ / ₁₆



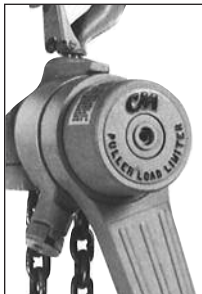
Shipyard hooks are available for use with the Series 653 in shipbuilding and metal fabrication industries. The upper shipyard hook replaces the standard hook and easily attaches to the body of the hoist. The lower hook is furnished with a chain block for direct attachment to the load of the hoist. Shipyard hooks are excellent for positioning steel plate and fixtures during welding applications.



Puller

The CM Puller is designed for heavy-duty construction and industrial applications. Used to pull, lift, drag or stretch, it features:

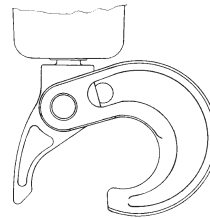
- Tough aluminum alloy construction and powder coat finish
- Weatherproof for outdoor service
- Simple construction with fewer parts for ease of maintenance and lower inventories
- Hoist alloy hardened steel link type load chain for strength, long wear life and flexibility
- Weatherized Weston-type automatic braking system for positive load control
- Easy, one-hand operation and control — only 58 pounds of pull required for $\frac{3}{4}$ ton model capacity
- Forged upper and lower hooks with latches standard
- Free wheeling for fast and easy attachment to load
- Upper and lower Latchlok hooks available for all capacities
- Optional Load Limiter protection device stops transmission of lever forces protecting against dangerous overload
- Optional anchor sling simplifies attachment to allow anchor hook to swivel in tight space applications ($\frac{3}{4}$ & $1\frac{1}{2}$ ton units only)
- Optional Load Sentry warns of overload condition
- Optional zinc-plated chain available
- Unlimited lift
- Lifetime warranty
- Metric rated
- Made in U.S.A.



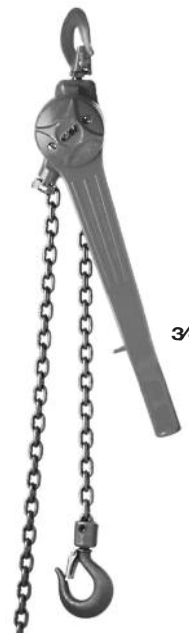
Optional Load Limiter



Optional anchor sling ($\frac{3}{4}$ and $1\frac{1}{2}$ ton)



Latchlok type hook



$\frac{3}{4}$ ton

Puller specifications - standard unit

Rated capacity* (tons)	Product code	Standard lift† (ft.)	Pull to lift full loads (lbs.)	Minimum distance between hooks (in.)	Lever length (in.)	Reeving	Hook throat openings (in.)	Approx. shipping weight (lbs.)
$\frac{3}{4}$	4043	5	58	$10\frac{3}{4}$	$21\frac{1}{4}$	1	$1\frac{1}{32}$	16
$1\frac{1}{2}$	4045	5	89	$14\frac{1}{4}$	$21\frac{1}{4}$	1	$1\frac{1}{8}$	26
3	4047	5	95	$16\frac{15}{16}$	$21\frac{1}{4}$	2	$1\frac{7}{32}$	38
6	4050	5	96	$21\frac{3}{8}$	$21\frac{1}{4}$	4	$1\frac{3}{4}$	73

† Can be supplied with lifts longer than 5 feet

Optional kits

Rated capacity* (tons)	Load Limiter		Load Sentry kit			Anchor sling kit	
	Product code	Approx. shipping weight (lbs.)	Product code	Lever length (in.)	Approx. shipping weight (lbs.)	Product code	Approx. shipping weight (lbs.)
$\frac{3}{4}$	4190	$8\frac{1}{4}$	4096	$23\frac{3}{4}$	$4\frac{1}{2}$	4071	2
$1\frac{1}{2}$	4191	$8\frac{1}{4}$	4098	$23\frac{3}{4}$	$4\frac{1}{2}$	4072	$4\frac{3}{4}$
3	4191	$8\frac{1}{4}$	4098	$23\frac{3}{4}$	$4\frac{1}{2}$	—	—
6	4191	$8\frac{1}{4}$	4098	$23\frac{3}{4}$	$4\frac{1}{2}$	—	—

Short handle puller

The CM Short Handle Puller is designed for easy operation, especially in hard to reach small spaces. It features:

- Easy one-hand operation
- Lever pull range is 45 to 62 pounds to lift full load
- Handle that rotates 360°, making it easy to operate in any position and in small spaces
- Tough yet lightweight aluminum construction and powder coat finish
- Hoist alloy hardened steel load chain for strength, long wear life and flexibility
- Free chaining for fast, easy attachment to load
- Forged upper and lower hooks with latches
- Simple construction with fewer parts for ease of maintenance
- Optional Load Limiter device protects against overload
- Lifetime warranty
- Made in U.S.A.



Specifications - Short handle puller

Rated capacity* (tons)	Product code	Standard lift (ft.)	Pull to lift full loads (lbs.)	Minimum distance between hooks (in.)	Lever length (in.)	Reeving	Hook throat openings (in.)	Approx. shipping weight (lbs.)
3/4	7300P	5	45	12 1/4	12	1	1 1/32	16
1 1/2	7320P	5	55	14 3/4	16 3/4	1	1 1/8	27
3	7330P	5	62	17 5/8	16 3/4	2	1 3/16	40

Specifications - Short handle puller without chain

Rated capacity* (tons)	Product code	Approx. shipping weight (lbs.)
3/4	7301P	12
1 1/2	7321P	21
3	7331P	28



Optional Load Limiter



Optional upper Latchlok Hook 3/4 – 1 1/2 ton



Optional anchor sling

Optional kits

	Product code	Approx. shipping weight (lbs.)	Rated capacity* (tons)
Load Limiter	7310P	1	3/4
	7311P	2	1 1/2-3
Anchor Sling	7309P	2	3/4
	7312P	3	1 1/2
Latchlok Hook	1001	2	3/4
	upper 1003	2	1 1/2
	1004	3	3
	lower 1001	2	3/4
	1003	2	1 1/2
	1005	3	3



Optional upper Latchlok Hook 3 ton

Rigger lever operated hoist

Designed for moderate commercial applications, this lightweight, compact tool stores easily and features:



- Flexible, non-binding CM load chain
- Forged upper and lower hooks with latches standard
- Free-wheeling chain for fast attachment
- Clover leaf chain guides and stripper seat chain in liftwheel reduce chance of jamming
- Unlimited lift by ordering any length chain
- Ratchet, driving pawl and holding pawl for load control
- One lever used to change from lifting to lowering and back to lifting
- Easy maintenance
- One-year warranty
- Metric rated
- Made in U.S.A.

WARNING

Overloading and improper use can result in injury.

To avoid injury:

- Do not exceed working load limit, load rating capacity
- Do not use to lift people or loads over people.
- use only alloy chain for overhead lifting.
- Read and follow all instructions.

Specifications

Rated capacity* (tons)	Product code	Standard lift† (ft.)	Reeving	Pull to lift full load (lbs.)	Minimum distance between hooks (in.)	Lever length (in.)	Hook throat openings (in.)	Shipping weight (lbs.)
3/4	2301	5	1	98	10 3/4	15 3/8	1 1/32	8 3/4
1 1/2	2302	5	2	98	12 1/2	15 3/8	1 1/32	13 3/4

† Can be supplied with lifts longer than 5 feet



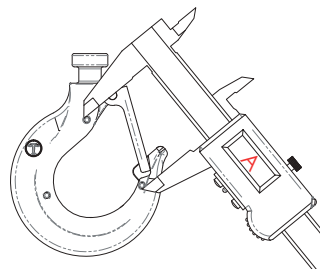
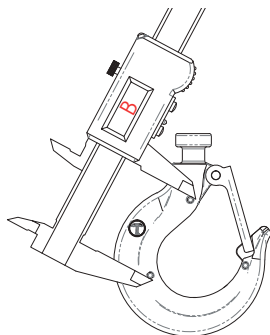
Tiger

Product Catalogue

Tiger hoisting, clamping, winching, and height safety ranges

V1.1 | 2021 | tigerlifting.com

Chain blocks and lever hoists overview



Hooks:

Our hooks incorporate our latest patented “EZ Check” feature which allows operators and inspection engineers to easily check to see if the hoist has been previously overloaded or shock loaded by simply measuring between the triangle of pop marks. With this system a quick check that the A and B measurements are the same (as shown in the diagram above) will indicate if the hooks have stretched: If $A \leq B$ the hook is fine; If $A > B$ the hook needs to be replaced.

They are drop forged and heat treated alloy hooks with heavy duty spring latch kits that fully engage into the recessed nose of the hook safeguarding the load when side loading of the latch occurs and offering maximum load security. The swivel hooks are attached to the yoke using high grade nuts & bolts allowing for complete inspection during any statutory inspection or written scheme maintenance. The entire hook fitting has been protected using corrosion protected or stainless steel fixtures and fittings.

Certification:

Each unit comes fully certified with a test certificate or an EC Declaration of Conformity stating compliance with the essential health and safety requirements of the Machinery Directive 2006/42/EC. This document clearly states the proof test at 1.5 times the rated capacity and the light load test of 2 % of the rated capacity, which exceeds the requirements of the current applicable standard. Finish: Tiger hoists are finished in our high quality powder coat paint offering excellent corrosion protection.

Load chains:

Tiger only use the highest quality alloy, calibrated load chains in Grade T or VH conforming to EN818-7 and ISO 16872. Our hoists are also available with stainless steel load chain.

Finish:

Tiger hoists are finished in our high quality powder coat paint offering excellent corrosion protection

Phosphor bronze bearings:

The Tin content increases the corrosion resistance and strength of the alloy, the Phosphorus increases the wear resistance and stiffness of the alloy. Needle Bearings have been proven to be unstable and unsuitable for offshore and marine environments.



Multi fall load sheave bearings:

We have incorporated DU® self-lubricating metal polymer bearings into our multi chain fall load sheaves. Their success lies in the unique composite bearing structure comprised of a steel backing for mechanical strength and a porous sintered bronze interlayer impregnated with PTFE and lead for high wear resistance and extremely low friction properties. This makes them ideally suited to hoists used subsea. They provide improved energy efficiency due to low friction of PTFE-enriched overlay and have an extended service life with a strong wear-resistant bronze layer. There is zero maintenance as self-lubricating bearings can operate perfectly dry but have greater reliability with lubricant to avoid potential shaft/bearing seizure under boundary conditions.



Performance:

Tiger chain blocks and lever hoists have undergone a full test program that includes:

Type Testing

- Ultimate strength test, at least 4 times the Working Load Limit (WLL)
- Continuous cyclic testing of over 1500 cycles
- Chain end anchorage test, passing at least 2.5 times the WLL without any restriction of the brake or gears

Test on every hoist

- Proof loading at 1.5 times the rated capacity
- Light load testing at maximum of 2% rated capacity

Mechanical efficiency:

All major components are heat treated and precisely machined to ensure long term smooth operation. High gear ratios keep physical effort to a minimum. Grease: High performance premium quality grease allows our hoists to be certified for use within a temperature range of -40°C to +80°C and can be customized to -60°C to +80°C, far exceeding the requirements of current standards.

Note: All such operations should be carried out in accordance with manufacturer's instructions and fully risk assessed by a competent person.

Cross hauling/fleeting operations:

Chain blocks and lever hoists fitted with our patent protected Quad Cam brake system have undergone a full and comprehensive testing regime that was drawn up by industry specialists and Tiger's mechanical engineers to provide evidence that these units can be safely used for fleeting and cross-hauling applications up to 45° without de-rating.

Brake efficiency:

Tiger hoists utilise an 8 point multi-start thread pinion shaft which doubles the lead distance compared with typical 4 point threaded shafts, increasing the brake efficiency by 100%. The rotational engagement of the brake works with the direction of the load sheave creating highly efficient braking ratios. With our fine manufacturing tolerances between the pinion shaft thread and critical brake components we have engineered out a known failure mode where contamination to the threads can lead to brake failure or slippage.



Globally patent protected



Quad cam brake design:

Our patented braking system feature has been designed by Tiger to enhance the safety of our lever hoists and chain blocks. This unique design ensures there is always permanent engagement of one of the pawls in the ratchet wheel. The reciprocating action always forces the pawl into the gear even without a pawl spring. This ensures the brake is always fully functional. With a conventional twin pawl brake mechanism (tear drop design) failure can occur if the pawl spring erodes, fractures or a shock load bounces the pawls out of the ratchet gear. Without the spring these systems rely on gravity keeping the pawls engaged and are flawed when the hoist is used at angles away from the vertical. Our quad cam pawl system is fitted to PROCB/SS20 chain blocks and PROLH/SS19 lever hoists.

Overload protection:

If the exact load can't be accurately determined then consider using hoists with slipping clutch overload protection. If the chain block or lever hoist is fitted with a slipping clutch overload mechanism then it is protected from overload. When the load limit is exceeded, the clutch in the chain wheel will slip, allowing the hand wheel to be rotated or handle to be operated whilst not engaging the gear box. This prevents further lifting of the load, though lowering is still possible. Load limiters can be fitted to PROCB/SS20 chain blocks and PROLH/SS19 lever hoists. Tiger chain blocks and lever hoists fitted with overload limiters are DNV GL Verification tested according to NORSOK R-002.



Additional adaptable fittings for PROCB/SS20 chain blocks and PROLH/SS19 lever hoist. To replace the top hook, bottom hook or both hooks



Clevis self-locking hook



Clevis shackle



Clevis sling hook



Clevis reeveable egg/master link



500kg lever hoist with clevis self locking hook



Spark resistant chain blocks and lever hoists



Ex II 2 GD c IIC T4 IIIC T135°C.

In many areas of industry, plants are operated with potentially explosive atmospheres. Due to the very high risk to personal and other assets, extremely strict guidelines and technical requirements are in place in explosive zones. Tiger Spark Resistant products are manufactured in accordance with ATEX Directive 2014/34/EU and the Machinery Directive 2006/42/EC. Our products are lightweight and durable with a proven track record.

Atex marking: Ex II 2 GD c IIC T4 IIIC T135°C. Suitable for gas and dust hazardous areas: Zone 1 & 2 (gas) and Zone 21 & 22 (dust).

We can supply chain blocks, lever hoists, beam clamps, trolleys and combination block and trolley units.



- Category II – high safety - Certification: Ex II 2 GD c IIC T4 IIIC T135°C
- Manufactured in accordance with ATEX Directive 2014/34/EU and Machinery Directive 2006/42/EC
- Developed for the use in potentially explosive atmospheres (ATEX zones)
- Special coated unit
- Different capacities and heights of lift to suit your specific requirements
- Available with stainless steel load chain
- Available with overload protection

Note: It is the responsibility of the user to identify which explosion proof classification is required, not the hoist manufacturer or supplier.

Lever hoists



All our lever hoists are manufactured from the highest quality materials precisely machined in our own factory, producing strong products with high resistance to impact damage. The hoists are finished in our high quality powder coat paint offering excellent corrosion protection. This all ensures they are ideal hoists for long-term use with less maintenance and servicing costs. The strong impact resistant gearbox housing, complete with close tolerance fit bearings that fully support the pinion shaft and planet gears, maintains smooth revolutions whilst under load. High gear ratios keep physical effort to a minimum; this provides maximum mechanical efficiency.

These units meet and exceed all the requirements within BS EN 13157: 2004+A1:2009, ANSI/ASME B30.21, AS1418.2 and SANS 1636.

Industrial TRLH



- Range from 750kg-6.0t
- Drop forged and heat treated alloy hooks
- Safe reliable twin pawl mechanical brake
- High quality alloy calibrated load chain
- Proof tested to 1.5 times rated capacity
- Guaranteed light load protection
- 1.5m height of lift as standard but can be chained to exact requirements
- Optional extra heavy duty chain end stop

Mini Series



- 250kg and 500kg units
- Safe and reliable twin pawl mechanical brake
- Rugged steel body construction
- Lightweight and portable
- 360 degree handle rotation
- Short stroke and low lever pulling effort
- Easy single-hand "freewheeling" operation
- Guaranteed light load protection
- Heat treated alloy steel triple spur gears and pinion shaft
- Operating temperature -40°C to +80°C (temperature range of -60°C to +80°C available as an option)
- Supplied with useful belt bags

Professional PROLH



UK Network Rail
PADS Certification

- Range from 800kg –20.0t
- Patented Quad Cam Pawl System
- DNV GL Verification tested according to NORSOK R-002
- Available with slipping clutch overload protection
- Available with forged clevis adaptor components
- Equipped with our unique torsion brake mechanism
- One piece construction pinion gear
- No requirement for preload tension to activate the ratchet mechanism
- Light load tested and certified at 2 % of the rated capacity
- Adaptable to use both inverted and horizontally
- Individual spares readily available
- Operating temperature -40°C to +80°C (temperature range of -60°C to +80°C available as an option)
- All major parts are heat treated and precisely machined
- Finished in high quality powder coat paint offering excellent corrosion protection
- Can be chained up to meet your exact requirements
- Phosphor bronze bearings
- Adjustable chain end anchorage available as an option
- Also available with stainless steel load chain or spark resistant hooks
- Available with load bearing swivel hooks to meet relevant standards i.e. AS 1418.2



SS19 and PROLH gearboxes contain one-piece pinion gears giving unprecedented strength in the harshest environments.



8 point multi-start thread pinion shaft on all our lever hoists and chain blocks- doubles the lead distance compared with typical 4 point threaded shafts, increasing the brake efficiency by 100%.



User friendly heavy duty end stops on our PROLH (and as an option on our TR7) make it easy for the operator to position the chain while "freewheeling". Most importantly this "end stop" will hold a load of at least 2.5 times the rated capacity when supporting the full load without restraint from other components e.g. brake or gearing.



Tiger Professional and Specialist lever hoists have our unique rotational inertia driven torsion brake system. The unique brake system guarantees a light load protection at 2% of the rated capacity.

Specialist SS19



- Range from 800kg –20.0t
- All the features of the ProLH as well as the following:
- Suitable for use in multi-immersion applications (see p25)
- Meets and exceeds IMCA DO28 Rev 2 guidance notes for offshore lever hoists
- New even harder-wearing paint finish in accordance with ISO12944 C5M
- Marine specific friction discs
- Full face gasket provides brake chamber protection from outside contamination
- Fully corrosion protected body and components to ensure reliable long-term use
- Over 1500 hours protection to ASTM B117 conditions
- Adjustable chain end anchorage specifically designed to the contours of the hoist
- New technology high performance grease

Download our technical sheet for more details:



Tiger have incorporated the ultimate safeguarding of a single point failure mode as described in IMCA DO28 Rev. 2 Note 7.2 by installing a specially designed adjustable chain end stop. Tiger have specifically profiled and manufactured this item to fit the contours of the SS hoist. This offers complete assurance to the operator, whilst maintaining the flexibility of "freewheeling" the chain to any required starting position.



The fluted chain stripper is specifically designed to guide the load chain safely onto or off the load sheave. The critical configuration of, and distance between, the chain stripper and the roller guide ensures alignment of the chain so it cannot snag or jam. This allows the safe use of the SS19/PROLH when used in any orientation.



The SS19 has bonded marine friction discs developed and designed for subsea use creating a highly efficient, reliable braking surface for use in the most extreme conditions.



Precisely engineered side plates and components protected by a full gasket give the SS19 brake chamber and contents maximum protection from external contamination. Phosphor bronze bearings are ideally suited to offshore marine environments.

Mini Series

Product Code	Capacity (tonne)	Effort (kg)	Dimensions (mm)							Load Chain		Standard HOL (m)	Weight @ std HOL (kg)	Weight for extra metre HOL (kg)
			A	B	C	D	E	F1	F2	Diameter (mm)	No. of falls			
LH-0025	0.25	30	60	168	245	85	66	22	31	4.0	1	1.5	2.0	0.3
LH-0050	0.50	30	89	168	280	113	70	24	38	4.0	1	1.5	3.0	0.3

Industrial TRLH

Product Code	Capacity (tonne)	Effort (kg)	Dimensions (mm)							Load Chain		Standard HOL (m)	Weight @ std HOL (kg)	Weight for extra metre HOL (kg)
			A	B	C	D	E	F1	F2	Diameter (mm)	No. of falls			
TRL-0075	0.75	22	128	240	275	158	99	24	38	6.3	1	1.5	7.5	0.9
TRL-0100	1.0	29	128	240	295	158	99	28	45	6.3	1	1.5	7.5	0.9
TRL-0150	1.5	26	154	360	320	172	104	34	51	7.1	1	1.5	10.0	1.1
TRL-0200	2.0	35	154	360	367	172	104	35	53	7.1	1	1.5	10.5	1.1
TRL-0300	3.0	38	182	360	400	195	108	36	56	10.0	1	1.5	18.0	2.2
TRL-0600	6.0	40	242	360	570	195	108	49	70	10.0	2	1.5	29.5	4.3

Professional PROLH

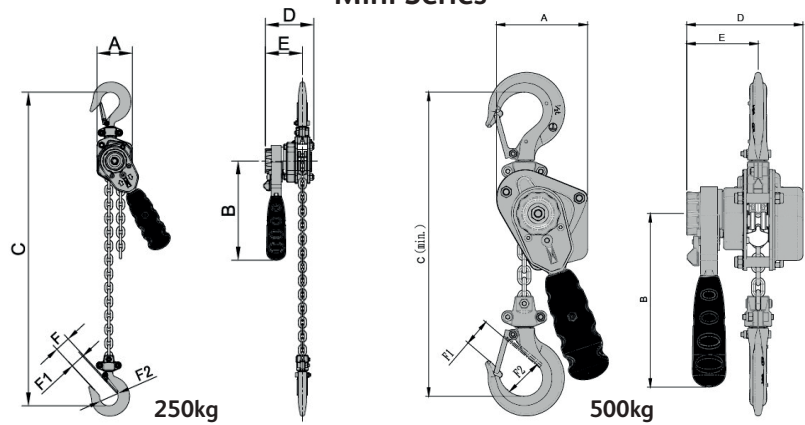
Product Code	Capacity (tonne)	Effort (kg)	Dimensions (mm)							Load Chain		Standard HOL (m)	Weight @ std HOL (kg)	Weight for extra metre HOL (kg)
			A	B	C	D	E	F1	F2	Diameter (mm)	No. of falls			
PLH-0080	0.8	23	128	240	295	158	99	28	45	6.3	1	1.5	7.5	0.9
PLH-0160	1.6	28	154	360	320	172	104	34	51	7.1	1	1.5	10.5	1.1
PLH-0320	3.2	40	182	360	400	195	108	36	56	10.0	1	1.5	18.0	2.2
PLH-0630	6.3	42	242	360	570	195	108	49	70	10.0	2	1.5	28.5	4.3
PLH-0800	8.0	38	379	360	580	195	108	51	78	10.0	3	1.5	43.0	6.5
PLH-1000	10.0	47	379	360	630	195	108	54	87	10.0	3	1.5	46.0	6.5
PLH-1500	15.0	44	566	360	840	195	108	59	81	10.0	5	1.5	112.0	10.8
PLH-2000	20.0	49	470	360	1050	250	160	81	110	10.0	6	1.5	156.0	13.1



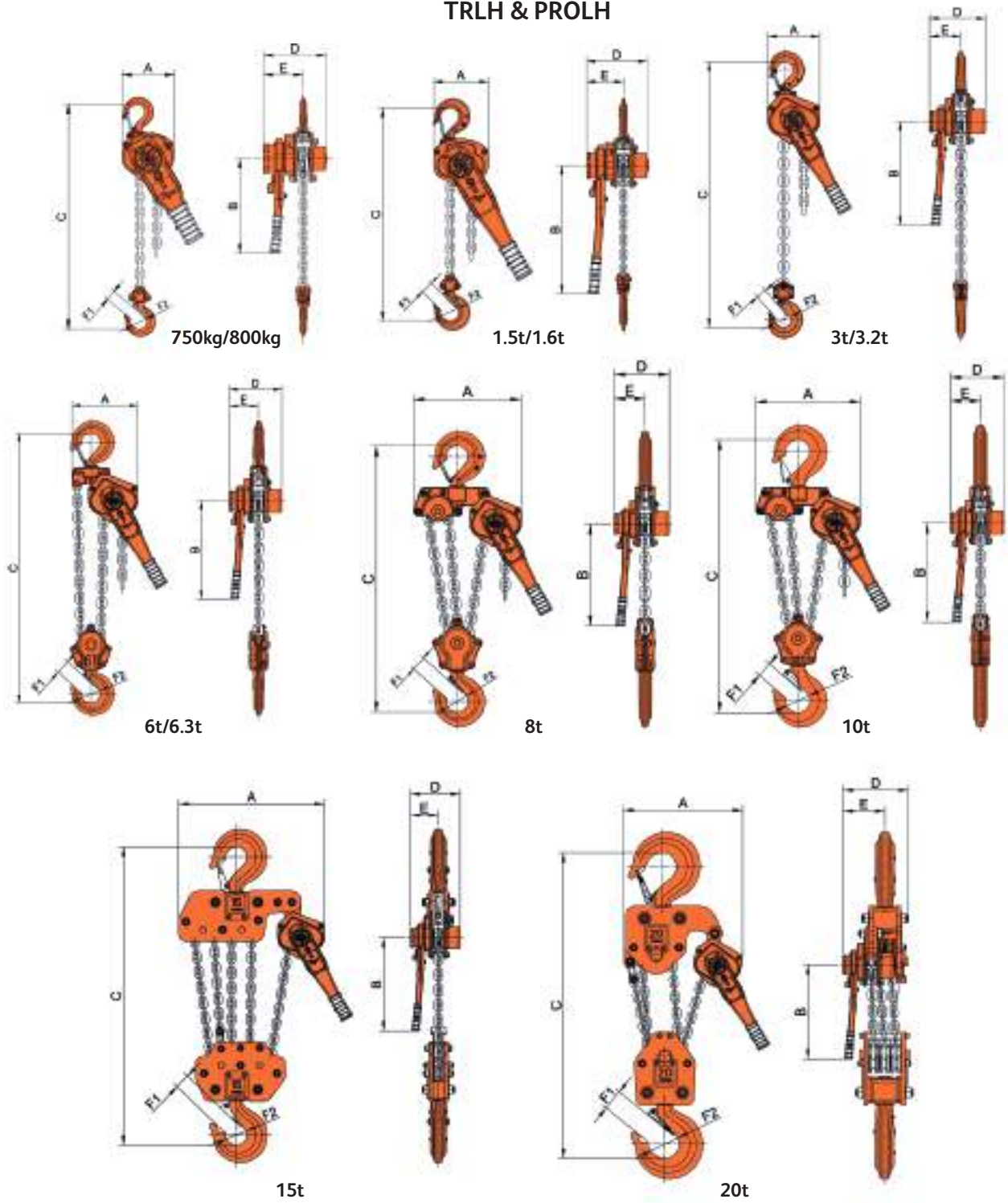
Available with clevis adaptor fittings (see page 16) and load limiters on request..



Mini Series

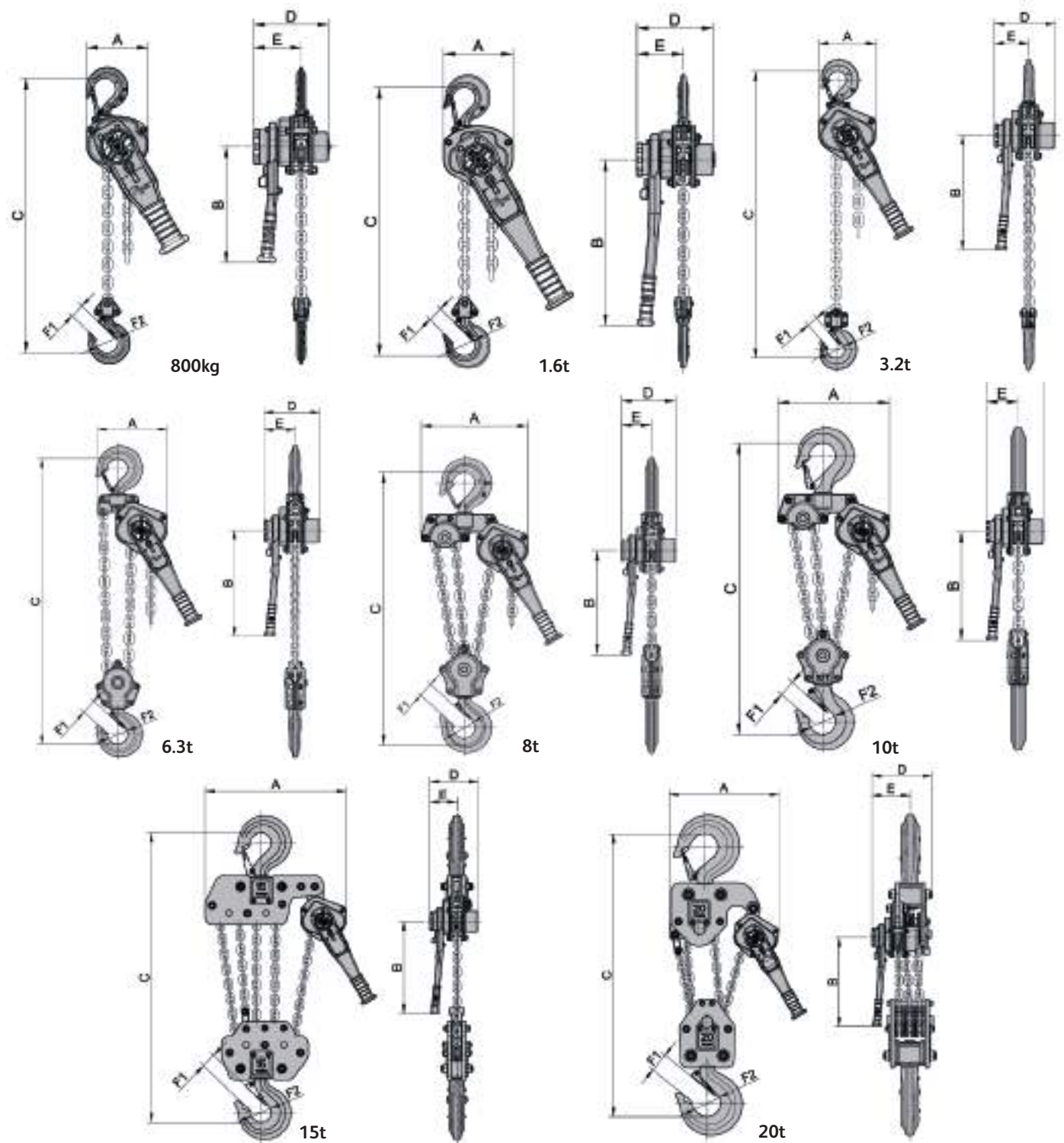


TRLH & PROLH



Specialist SS19

Product Code	Capacity (tonne)	Effort (kg)	Dimensions (mm)							Load Chain		Standard HOL (m)	Weight @ std HOL (kg)	Weight for extra metre HOL (kg)
			A	B	C	D	E	F1	F2	Diameter (mm)	No. of falls			
SS-0080	0.8	23	128	243	295	158	99	28	45	6.3	1	3	9.0	0.9
SS-0160	1.6	28	154	370	320	172	104	34	51	7.1	1	3	12.5	1.1
SS-0320	3.2	40	182	370	400	195	108	36	56	10.0	1	3	22.0	2.2
SS-0630	6.3	42	242	370	570	195	108	49	70	10.0	2	3	34.5	4.3
SS-0800	8.0	38	379	370	580	195	108	51	78	10.0	3	3	52.0	6.5
SS-1000	10.0	47	379	370	630	195	108	54	87	10.0	3	3	55.0	6.5
SS-1500	15.0	44	566	370	840	195	108	59	81	10.0	5	3	128.0	10.8
SS-2000	20.0	49	470	370	1050	250	160	81	110	10.0	6	3	174.0	13.1



Subsea & multi-immersion



HOISTING

After successfully completing a series of third party verification and testing programmes, our latest being a multi-immersion programme of tests over 54 days developed and agreed by industry professionals and independently carried out and verified at ORE Catapult in the UK (a leading technology innovation and research centre for offshore renewable energy), we can state that the Tiger SS range of lever hoists and chain blocks are suitable for use in multi-immersion applications providing the timescales, pre-use testing, maintenance and service requirements as detailed in Tiger document Multi-Immersion Instruction Manual are strictly followed.



We used a subsea dock including a replica seabed



Saltwater was pumped in from the Blyth Estuary (North Sea)



We have also carried out intensive internal testing to ensure we achieved over 1500 hours protection to ASTM B117 conditions. This ensures the performance of the SS ranges has been fully challenged in natural environments similar to the conditions they will be exposed to.

Tiger SS19 and SS20 hoists are completed with a two-stage finish combining electroplating as a base layer and then finished with a very hard-wearing sand texture powder coating paint which offers excellent corrosion protection and high impact protection ideally suited for multi-immersion applications. Paint finish in accordance with ISO12944 C5M.

We use new-technology, wide temperature range, high performance premium grease. It contains no heavy metals or other harmful or environmentally undesirable additives. This selection achieves the correct viscosity to prevent the grease migrating from the area of application. Even when mixed with 50 % high water absorption it has excellent adhesion and no sign of breakdown after 100,000 strokes and remains unchanged with no slump or breakdown unlike other "premium" greases. Load carrying ability, mechanical stability, corrosion protection, water resistance and increased operating temperature ranges are all improved by the using only the very best lubricant.

Standard operating temperature range -40°C to +80°C - far exceeding the requirements of current standards. Ultra-low operating temperature units also available to -60°C.



We have incorporated DU® self-lubricating metal polymer bearings into our multi chain fall load sheaves. Their high wear resistance and extremely low friction properties makes them ideally suited to hoists used subsea.



Our latest forged latches incorporate a new wing design making them easier to operate in the operation area (e.g. wet gloved hands for offshore or subsea use).



Tiger only utilise phosphor bronze bearings on hoists. The tin content increases the corrosion resistance and strength of the alloy, the phosphorus increases the wear resistance and stiffness of the alloy. This makes them ideally suited for subsea environments.



ALL MATERIAL HANDLING
Lifting Equipment

Product Catalog #20





X5 Series Hoist

X5 Lever Chain Hoist

¾ to 10 US Ton Capacity

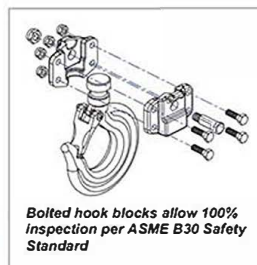
¾ to 9 Metric ton Capacity

FEATURES

- ▶ Brake sets and operates without minimum preload.
- ▶ 100% inspection of hooks per ASME B30 and easy hook replacement is permitted by bolted hook blocks.
- ▶ All steel body for durable strength in a smaller size.
- ▶ Powder coated and plated external parts to resist corrosion.
- ▶ Caged roller bearings on chain wheels for highest efficiency.
- ▶ Laser etched stainless steel nameplates with required OSHA/ASME B30 warnings, model and stamped serial number.
- ▶ Weston style mechanical load brake is self-adjusting with double pawls and is surrounded by an enclosure to limit contamination.
- ▶ Fused brake disc cannot crack, shatter, or absorb moisture.
- ▶ Grade 100 zinc plated load chain made in Japan or EU.
- ▶ Optional overload protection is a one directional slip clutch that only slips in the lift direction.
- ▶ Ambient Temperature range: -40° to +140° F (-40° to +60° C).
- ▶ Meets: OSHA, ASME B30.21 and ASME HST-3. In addition, the ultimate strength is 5 times Working Load Limit (WLL), meeting structural requirement of ASME HST-3 Appendix A, for use in marine and other applications as specified by US Department of Defense (DOD).
- ▶ ATEX rated Ex II 3 GD c IIB 60° C X for limited use in hazardous environments.
- ▶ Final assembly and load test in the USA from imported components. Supplied with certificate by serial number.
- ▶ Standard lifts are 5, 10, 15 and 20 ft. Custom lifts are also available.



Oversized laser etched stainless steel nameplate with required ASME/OSHA WARNINGS



Bolted hook blocks allow 100% inspection per ASME B30 Safety Standard

X5S ADDITIONAL SUBSEA MODEL FEATURES

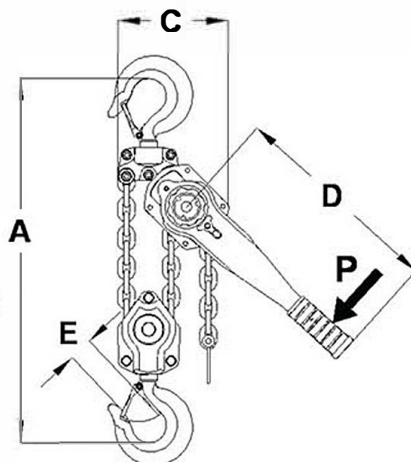
- ▶ SubSea models for use submerged under water.
- ▶ Brake is enclosed in a gasket sealed area along with bolted-in stainless dual springs and pawls.
- ▶ Meets IMCA D028 standard (UK sub-sea lever hoist standard).
- ▶ Extensive coatings, finishes and materials to resist corrosion.
- ▶ Fitted with an adjustable chain stop to meet local codes for use under water.



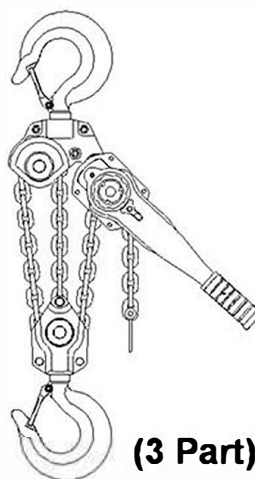
Adjustable Stop



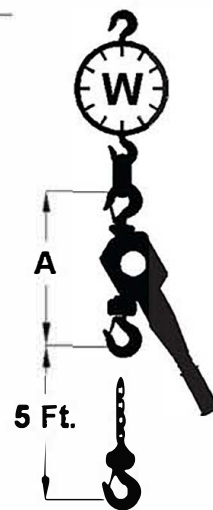
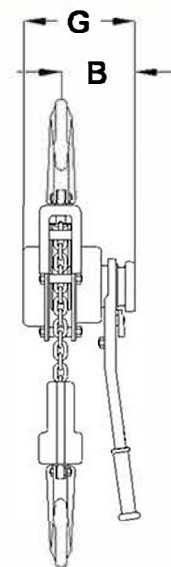
(1 Part)
3/4 - 3.5 T



(2 Part)
6 - 7 T



(3 Part)
9 - 10 T



Working Load Limit (WLL)			Specifications (inches and pounds)										
Base Model	WLL US Ton	kg	Chain Grade 100	Parts	A" Min.	B"	C"	D"	E" with Latch	G"	P (lbs)	W (lbs) /Ft Lift	W (1) (lbs) 5' Lift
X5L02250	3/4 to 1 1/8	1020	C5.6H17Z	1	11.8	3.8	4.7	9.6	1.1	5.7	66	0.5	13.5
X5SL02250													
X5L03528	1 1/2 to 1 3/4	1600	C7.1H20Z	1	13.2	3.9	5.0	10.4	1.2	6.5	68	0.7	18.0
X5SL03528													
X5L07056	3 to 3 1/2	3200	C10H28Z	1	15.6	4.5	6.3	16.3	1.4	7.7	84	1.5	36.5
X5SL07056													
X5L14000	6 to 7	6350	C10H28Z	2	21.3	4.5	8.6	16.3	1.8	7.7	86	3.0	55.0
X5SL14000													
X5L20000	9 to 10	9070	C10H28Z	3	26.8	4.5	11.7	16.3	2.1	7.7	88	4.5	90.0
X5SL20000													

Order by descriptive model number

Example: **X5L07056-05VS**

Base Model

Options (Not available for SubSea models)

V = Overload Protection, one-way operation will lower any load

S = Shipyard Hooks includes top and bottom (Available for 1 1/4 & 3 1/2 T)

L = Self-Locking Hooks includes top and bottom (Available for 1 1/8, 1 3/4 & 3 1/2 T)



S option



L option



Extender Kit
Order Separately



1 1/8 T

1 3/4 T

3 1/2 T

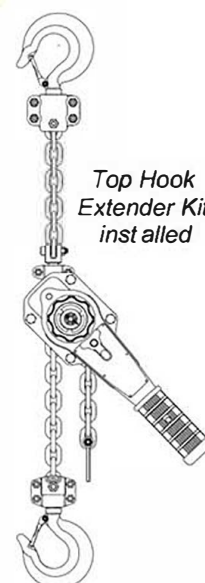


(Chain and Hook not included)

HOOK- EXT02250

HOOK- EXT03528

HOOK- EXT07056



Top Hook
Extender Kit
inst alled



Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.



LA Series Hoist

Lever Chain Hoist

3/4 to 10 US Ton Capacity

3/4 to 9 Metric ton Capacity

FEATURES

- ▶ Dual rated with more choices of capacity.
- ▶ Hook blocks are bolted, allowing easy disassembly for complete hook inspection.
- ▶ Lightweight and durable all steel construction with powder coated finish and plated external components to resist corrosion.
- ▶ Premium USA Made Grade 80 Zinc plated alloy load chain.
- ▶ Weston style mechanical load brake is self-adjusting with double pawls and is surrounded by an enclosure to limit contamination.
- ▶ Horseshoe chain guard allows operation in any orientation.
- ▶ Optional Overload Protection is a one directional clutch that will only slip in the lift direction.
- ▶ Meets: ASME/ANSI B30.21, OSHA, NASA-STD-8719.9
- ▶ ATEX rated Ex II 3 GD c IIB 54° C X for limited use in hazardous environments
- ▶ Long-life fused brake friction disc will not crack, shatter, or absorb moisture



Horseshoe Shape Chain Guide



Oversized laser etched stainless steel nameplate with required ASME/OSHA WARNINGS



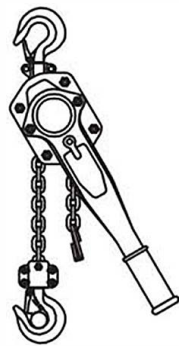
US Ton	Metric ton
3/4T	0.8t
1T	1.0t
1 3/4T	1.6t
2 1/4T	2.0t
2 3/4T	2.5t
3 1/2T	3.2t
7T	6.3t
10T	9.0t



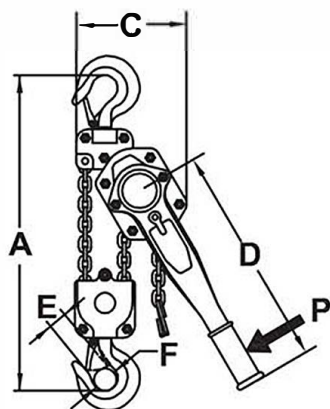
Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.

Final assembly and re-testing in the USA

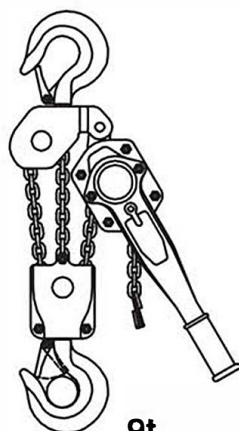




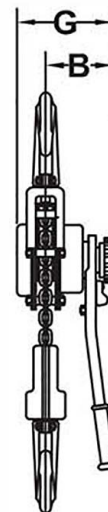
3/4-3.2t
(1 Part)



6.3t
(2 Part)



9t
(3 Part)



Working Load Limit (WLL)

Specifications (inches and pounds)

Base Model	US Ton	Metric ton	*Min. Load		Chain	Parts	A"	B"	C"	D"	E"	F"	G"	Effort P (Lbs)	W (Lbs) / ft Lift	W (1) (Lbs) 5 ft Lift
LA008	¾+	0.8	44	20	C6X18Z	1	13.4	3.9	5.2	10.5	1.02	0.98	6.2	47	0.54	14.9
LA010	1+	1	44	20	C6X18Z	1	13.4	3.9	5.2	12.1	1.02	0.98	6.2	51	0.54	15.3
LA016	1 ¾	1.6	70	32	C7X21Z	1	15.7	4.4	5.7	14	1.14	1.18	7.2	66	0.74	24.1
LA020	2 ¼	2	88	40	C8X24Z	1	18.1	4.6	7.6	14	1.38	1.3	8.1	66	0.92	36.2
LA025	2 ¾	2.5	110	50	C9X27Z	1	19.1	4.6	7.6	14	1.38	1.3	8.1	74	1.25	37.9
LA032	3 ½	3.2	141	64	C10X28Z	1	20.5	4.8	7.8	16.5	1.46	1.42	8.3	72	1.50	44.8
LA063	7	6.3	278	126	C10X28Z	2	25.2	4.8	9.1	16.5	1.69	1.69	8.3	74	3.00	66.3
LA090	10	9	396	180	C10X28Z	3	28.7	4.8	13.3	16.5	1.85	1.73	8.3	76	4.50	110.6

(1) Shipping weight can be estimated by adding approx. 15% to "W" hanging weight.

* **Minimum Load** = Load required to activate the automatic load brake. The brake will stop a load even if the operator leaves the unit in free chain mode or in neutral. This is a similar reaction to a seatbelt.

Standard Lift = 5, 10, 15, and 20 ft. Custom lifts are available.

Option: V = Overload Protection, one-way operation that will lower any load.

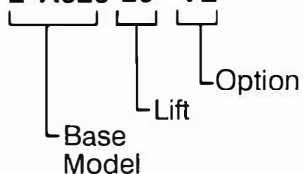
L = Self-Locking Hook Option

S = Shipyard Hooks. Available for 1.6t & 3.2t only.

Custom lifts

How to Order, by descriptive model number

Example: Model L A020 20 - VL



Fits into Standard Hook Block



S option

L option



Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.

Chain Hoists

1/2 to 10 Metric ton Capacity

FEATURES (for both lever and hand chain hoists)

- ▶ Low Headroom, Light weight and durable formed steel enclosure protects from contamination and can be quickly disassembled for fast and easy maintenance. Powder coating and plated finishes on exposed components provide corrosion resistance, allowing the hoist to stay in service longer.
- ▶ Internal parts are heat treated for strength and reliability with fully enclosed gearing to protect from contamination. All gears and shafts run on caged roller bearings, including the lift wheel which is machined for smooth operation and long life.
- ▶ Weston style mechanical load brake is self adjusting with double pawls and is surrounded by an enclosure to limit contamination.
- ▶ Drop forged Alloy hooks are designed to stretch when overloaded before chain failure and equipped with a heavy duty cast steel safety latch, attached by a bolt and lock nut. The top hook mounting articulates to aid in rigging and help ensure straight line loading between hooks.
- ▶ Grade 80 Alloy load chain with zinc plating.
- ▶ Each hoist is proof tested matching the individual serial number stamped on the nameplate. An operation and spare parts manual is also supplied in English, French and Spanish.
- ▶ Meets ASME B30, OSHA and NASA-STD-8719.9
- ▶ Unchained units and bulk chain are available for customers to make their own custom lifts if assembled and tested per ASME B30 before being placed into service.
- ▶ Spare parts are stocked in USA.



S Option:



Shipyard Hook



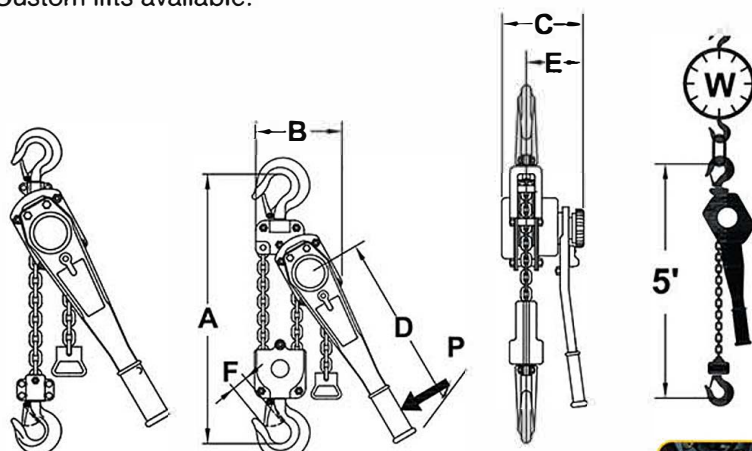
Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.

Lever Chain Hoist

¾ to 6 Metric ton Capacity

FEATURES

- ▶ Easy One Hand Operation and Set up - When the selector is in neutral and no load is present then free chain mode is entered. The chain can be pulled through the hoist by a pull handle for easy set up and load attachment before engaging the selector for pulling. The pull handle also acts as a chain stop. Short ratchet strokes of a non-slip rubber grip lever and the 360 degree rotation requires minimal space for operation. The pull force to operate may be 20 to 30% less than others.
- ▶ Operation in any orientation is assured by a fully enclosed horseshoe shape chain guard keeping the load chain in contact with the lift wheel.
- ▶ Available in lifts of 5, 10 and 20 feet zinc plated chain.
- ▶ Custom lifts available.



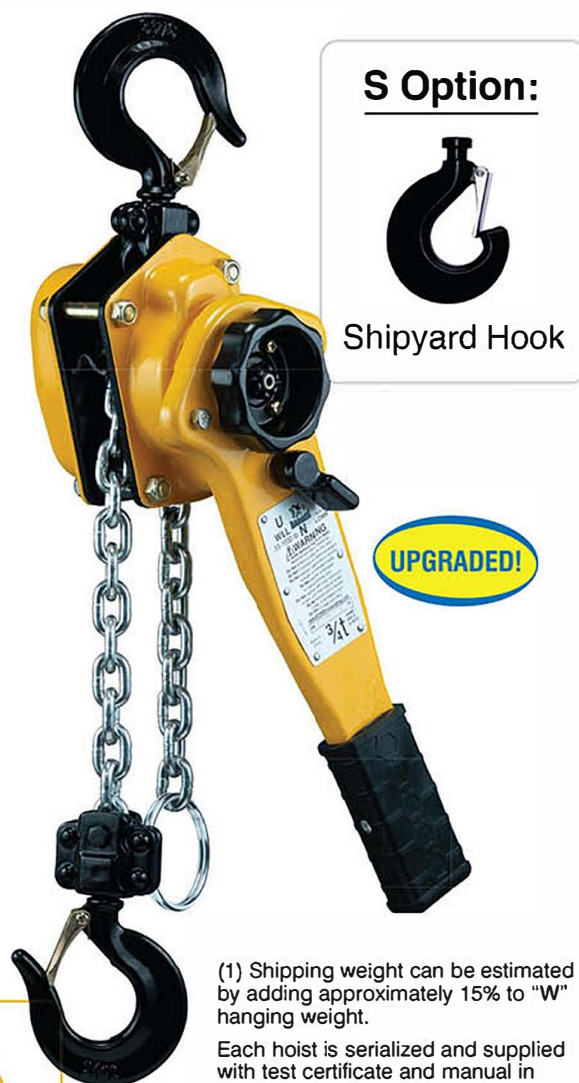
(1 part)
0.75, 1.5 & 3t

(2 part)
6t

Option: S = Shipyard Hooks, Available for 1.5t and 3t only

How to Order: By descriptive model number

Example: Model LC030-20S = 3t lever hoist with 20' lift shipyard hooks



(1) Shipping weight can be estimated by adding approximately 15% to "W" hanging weight.

Each hoist is serialized and supplied with test certificate and manual in English, French and Spanish.



Oversized laser etched stainless steel nameplate with required ASME/OSHA WARNINGS

SPECIFICATIONS AND DIMENSIONS (inches and pounds unless noted)

Base Model#	WLL Metric ton	Weight (lbs) Lift/FT	Weight (lbs) 5 Ft Lift	Chain# & Size (mm)	Effort P (lbs.) @ WLL	Falls	A	B	C	D	E	F with latch
LC008	0.75t (1650 lbs)	0.54	17	C6X18ZC	31	1	12.6	5.3	5.8	11.4	3.5	0.94
LC015	1.5t (3300 lbs)	0.92	26	C8X24ZC	54	1	15.0	6.4	6.9	16.5	4.0	1.14
LC030	3.0t (6600 lbs)	1.48	46	C10X30ZC	72	1	18.9	8.3	7.7	16.5	4.3	1.42
LC060	6.0t (13200 lbs)	2.96	70	C10X30ZC	76	2	24.4	10.0	7.7	16.5	4.3	1.81



Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.



Mini Hoists

Mini Chain Hoist 1/4 & 1/2 Metric ton Capacity

FEATURES

- ▶ Planetary gearing is used to minimize operator effort and to reduce stress on the brake components.
- ▶ Small size allows operation and storage in tight places.
- ▶ Lightweight and durable all steel construction with plated frame components and durable chrome exterior finish to resist corrosion.
- ▶ Grade 80 Zinc plated alloy load chain.
- ▶ Sleeve bearings used to maximize serviceability and provide good efficiency.
- ▶ Self-adjusting pawl and Weston style mechanical load brake ensures positive load control.
- ▶ Meets: ASME/ANSI B30, OSHA, NASA-STD-8719.9.
- ▶ ATEX rated Ex II 3 GD c IIB 54° C X for limited use in hazardous environments.



How to Specify Lever Model:

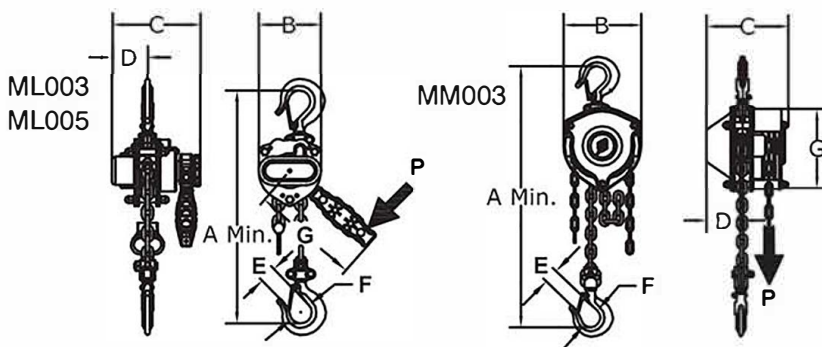
Example ML005-20
(1/2t Capacity, 20 ft lift)

Model	Lift
ML005	20

How to Specify Hand Chain Model:

Example MM003-20-18
(1/4t Capacity, 20 ft lift and 18 ft hand chain drop)

Model	Lift	Drop
MM003	20	18



M & L Series Specifications and Dimensions (inches and pounds)

Model	Working Load Limit (WLL)		Chain	A Min.	B	C	D	E	F	G	Effort P (lbs)	W (lbs) 5' lift
	Maximum	*Minimum										
MM003	1/4t / 550 lbs	0 lbs	C4X12Z	9.1	3.9	4.3	1.9	0.6	1	4.3	26	7.7
ML003	1/4t / 550 lbs	33 lbs	C4X12Z	8.5	3	4.6	1.6	0.6	1	6.2	28	4.9
ML005	1/2t / 1100 lbs	40 lbs	C5X15Z	10	3.5	5.3	2	0.8	1.3	7.1	40	7.7

* **Lever Min. capacity** is the load required to set the automatic safety brake. The brake will stop a load even if the operator leaves the unit in free chain mode and in neutral, in a similar way as a seat belt reacts.

Standard lift is 5 & 10 ft for Lever and 8 ft for Hand Chain Hoist models. All hoists are load tested at 125% Capacity.



Before using these products; read and follow instructions in the operator's manual. Never exceed capacity. Do not use for lifting people or loads over people.